Parallel lines

1

a Their gradients are the same.

b All graphs have the same angle of inclination or gradient.

2

a Yes **b** Yes **c** No **d** Yes

e No **f** No **g** No **h** No

3

a Yes **b** No **c** Yes **d** Yes

e No **f** No **g** No **h** No

4

a No **b** No **c** Yes **d** Yes

5

a Yes **b** No **c** No **d** Yes

6

a $-\frac{5}{8}$ **b** $-\frac{5}{8}$ **c** Yes

7 $y = 2x + 4$

8 $y = 2x + 4$

9

a $y = 2$ **b** $x = -7$ **c** $y = -3x - 4$ **d** $y = 8x + 5$

e $y = -2x - 5$

10

a $b = -\frac{5}{3}$ **b** $b = \frac{2}{5}$

11

a $-\frac{2}{5}$ **b** $y = -\frac{2x}{5} + 1$

12

a -1 **b** $y = -x + 18$

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13

a -5 **b** $y = -5x + 40$

Perpendicular lines

14 Two lines that intersect each other at an angle of 90°.

15 -1

16

a No **b** No **c** Yes **d** Yes

e No **f** Yes **g** Yes **h** No

17

a $m = -\frac{36}{5}$ **b** $m = 10$

18

a $-\frac{1}{6}$ **b** $-\frac{1}{6}$ **c** -8 **d** 1

19

a $-\frac{8}{7}$ **b** $\frac{8}{7}$ **c** No

20

a $x = -8$ **b** $y = -8$ **c** $y = 2x + 6$ **d** $y = -\frac{x}{6} + 10$

e $y = -\frac{x}{6} - 4$

21

a $-\frac{1}{5}$ **b** $y = -\frac{x}{5} + 1$

22

a $\frac{-1}{10}$ **b** $(5, 10)$

c $x + 10y - 105 = 0$

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Applications

23

a -3 **b** -3 **c** Yes

24

a $\frac{1}{5}$ **b** $\frac{1}{5}$ **c** Yes **d** Yes

e Parallelogram

25

a

i 1 **ii** -1

b Yes, because their gradients multiply to -1.

26

a

i $m = -1$ **ii** $m = 1$ **iii** $m = -1$ **iv** $m = 1$

b Square

27

a Yes **b** No **c** Yes

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